Halima Abdirizak Mohamed

Backend Engineering | Software Engineering | Application Security
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SUMMARY

Informatics student specializing in cybersecurity, with backend engineering experience building payment systems, usage-based billing, and multi-tenant APIs. Experienced in Python, FastAPI, and PostgreSQL, with a focus on secure authentication, data isolation, and automated testing. Also builds full-stack applications using React and TypeScript.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS
Backend and APIs: FastAPI, Flask, Node.js, Express, REST APIs, Pydantic, SQLAlchemy, Alembic
Frontend: React, Vite, Tailwind CSS
Databases: PostgreSQL, MongoDB, SQLite
Security and Infrastructure: Authentication, authorization, API security, secure data handling, OWASP Top 10, SafeLine WAF, Docker, Linux,
Tools: Git, GitHub, GitHub Actions, pytest, Vites

EXPERIENCE

System Intern, E-Technology Program | *President University, Cikarang, Indonesia* Sep 2026 – Present

- Administer and maintain Moodle course content for online classes, ensuring learning materials are organized and accessible.
- Developed and maintain a separate backup dashboard to support course content backup and recovery, providing an additional layer of protection for the program's learning materials.

Backend AI Engineering Intern | *FlyRank AI, Remote* Jul 2026 – Sep 2026

- Implemented Midtrans payment webhooks in FastAPI that verify HMAC-SHA512 signatures and use idempotency keys to prevent duplicate charges.
- Built usage metering, subscription quota enforcement and multi-tenant data isolation for Free and Pro plans with PostgreSQL, SQLAlchemy and Alembic, storing costs as integer cents.
- Wrote 68 automated pytest tests covering duplicate requests, plan limits, payment webhooks and tenant isolation.

PROJECTS

Dependency Compass Vulnerability triage tool that shows developers which dependencies to fix first
Python, FastAPI, OSV, CISA KEV | github.com/hal-imaxabdi/Dependency-compass

- Parses requirements.txt, package.json and package-lock.json files, checks each package against the OSV database, and ranks findings by CISA KEV status, severity and available fixes.
- Treats uploaded files as untrusted input with size, nesting and length limits, reports dependencies it could not verify, and is covered by 100+ pytest tests.

Hookline Chrome extension that flags phishing links in Gmail and explains why
TypeScript, React, Manifest V3, FastAPI | github.com/hal-imaxabdi/hookline

- Detects lookalike domains, homoglyphs, raw IP links and reply-to mismatches, with a FastAPI backend that checks domain age, registrar and SSL data through RDAP and DNS.
- Evaluated on 50,000 balanced samples: 76.1% precision and 40.5% recall, with limitations documented.

Slotwise Full-stack appointment booking platform for small businesses
React, TypeScript, FastAPI, PostgreSQL, Alembic | github.com/hal-imaxabdi/slotwise

- Prevented double bookings at the database level with a PostgreSQL exclusion constraint, and secured the API with HttpOnly cookie sessions, CSRF tokens, RBAC, login throttling and Argon2 hashing.
- Wrote 400+ backend tests in pytest and 280+ frontend tests in Vitest and React Testing Library.

SEMIS Secure HR management system mapped to the OWASP Top 10
Node.js, Express, React, MongoDB | github.com/hal-imaxabdi/SEMIS

- Built employee records, attendance, payroll and leave approval workflows with separate admin, HR and employee portals.
- Implemented email OTP MFA, JWT sessions, RBAC, bcrypt hashing, AES-256 encryption of salary and national ID fields, rate limiting and an append-only audit log.

EDUCATION AND CERTIFICATIONS

B.Sc. Informatics, Cybersecurity Concentration | President University | CGPA 3.73 / 4.00 Expected 2027

Certifications: Google Cybersecurity Professional Certificate, GitHub Foundations (GH-900), Google AI Professional Certificate, AWS Security Fundamentals (training)